

AWS EC2 Security Groups

A Security Group (SG) is a **virtual firewall** in AWS that controls inbound and outbound traffic for an EC2 instance. It decides who can access the instance and on which port.

Every EC2 instance must be associated with at least one security group.

SG Controls

- **Inbound traffic:** Who can come into the EC2 instance.
- **Outbound traffic:** Where the EC2 instance can go out (connect to).

Key Characteristics (Very Important)

Security Groups operate under a specific set of rules:

- Works at the EC2 instance level (not the subnet level).
- **Stateful firewall:** If inbound traffic is allowed, the response (outbound) traffic is automatically allowed back without needing a separate outbound rule.
- Supports only **ALLOW** rules; there are no DENY rules. If traffic is not explicitly allowed, it is automatically blocked.
- **Default inbound:** Deny all.
- **Default outbound:** Allow all.
- One EC2 instance can have multiple Security Groups attached.
- The same Security Group can be attached to multiple EC2 instances.

Common Ports

It is important to be familiar with the following common ports:

Port	Protocol/Service	Purpose
22	SSH	Linux access (Secure Shell)
3389	RDP	Windows access (Remote Desktop Protocol)
80	HTTP	Standard web traffic (non-secure)
443	HTTPS	Secure web traffic
3306	MySQL	Database connection
5432	PostgreSQL	Database connection

Simple Example

If SSH (port 22) is allowed only from my laptop IP address, only I can connect to the EC2 instance. If I remove that rule, even though the EC2 is running, I cannot access it.



Real-Life Analogy

This analogy helps simplify the concept:

- **EC2 instance** = House
- **Security Group** = Watchman at the door
- **Ports** = Doors of the house
- **IP address** = Visitor address

Result: Only visitors allowed by the watchman can enter.

One-Line:

Security Group is a **stateful virtual firewall** that controls inbound and outbound traffic for EC2 instances.

AWS Elastic Load Balancing (ELB)

A Load Balancer in AWS is a service that automatically distributes incoming traffic across multiple EC2 instances. Its main purpose is to ensure high availability, fault tolerance, and better performance of applications.

Instead of all users hitting one server, the load balancer spreads the load evenly. Load balancers sit in front of EC2 instances and act as a single entry point for users. Even if one EC2 instance fails, the load balancer sends traffic only to healthy instances, keeping the application running without interruption.

What Load Balancer Does

- Distributes traffic across multiple EC2 instances
- Performs health checks
- Improves availability & reliability
- Handles traffic spikes
- Provides single DNS endpoint for users

Types of Load Balancers (CANG)

Type	Layer	Key Feature	Common Use Case
Application Load Balancer (ALB)	Layer 7 (HTTP/HTTPS)	Supports path-based and host-based routing	Web apps, microservices
Network Load Balancer (NLB)	Layer 4 (TCP/UDP)	Very high performance, low latency	Extreme performance applications
Classic Load Balancer (CLB)	Layer 4/7	Old type	Mostly legacy (exam awareness only)

Key Characteristics

- Works at request/connection level
- Performs health checks on EC2
- Routes traffic only to healthy targets
- Can work across multiple Availability Zones
- Integrates with Auto Scaling
- Supports HTTPS (SSL/TLS termination)

Real-Life Analogy (Students remember this)

- Load Balancer = Traffic police 🚦
- EC2 instances = Multiple roads
- Traffic police sends cars to roads that are free and open

Simple Example

"If 1,000 users open a website, the load balancer sends requests to multiple EC2 servers instead of overloading one server." If one server goes down, users don’t even notice.

Common Use Case

- Websites
- E-commerce apps
- APIs
- High-traffic applications

One-Line :

A Load Balancer distributes incoming traffic across multiple targets to ensure high availability and fault tolerance.